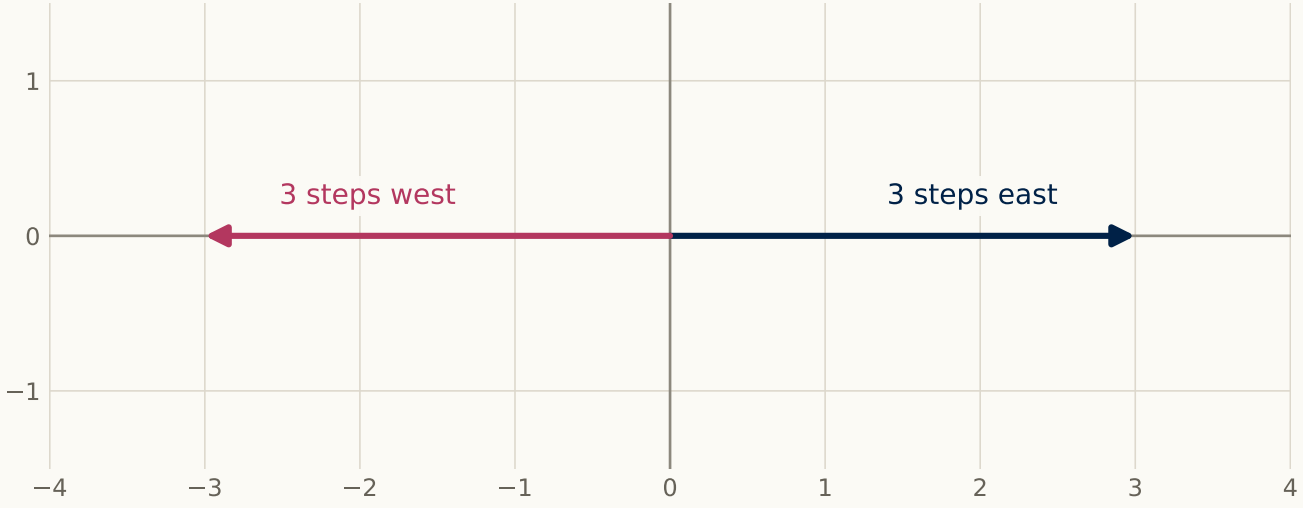Equal lengths, opposite journeys